

Day-5
Registers and Counters - Assignment

- 1) Design a counter with T-flipflops whose count sequence is 0, 2, 4, 5, 3, 1, 0, ...

⇒

Let,

Current state

Next state

require input

$Q_2 \quad Q_1 \quad Q_0$

$Q_2 \quad Q_1 \quad Q_0$

$T_2 \quad T_1 \quad T_0$

0 0 0

0 1 0

0 1 0

0 0 1

0 0 0

0 0 1

0 1 0

1 0 0

1 1 0

0 1 1

0 0 1

0 1 0

1 0 0

1 0 1

0 0 1

1 0 1

0 1 1

0 0 1

1 1 0

X X X

1 1 0

1 1 1

X X X

1 1 0

X X X

X X X

$T_{Q_0} =$

	$Q_1 Q_0$	00	01	11	10
Q_2	$\bar{Q}_2$	0 ₀	1 ₁	0 ₃	0 ₂
Q_2	Q_2	1 ₄	0 ₅	X ₇	X ₆

$$T_2 = Q_2 \bar{Q}_0 + \bar{Q}_2 \bar{Q}_1 Q_0$$

$T_1 =$

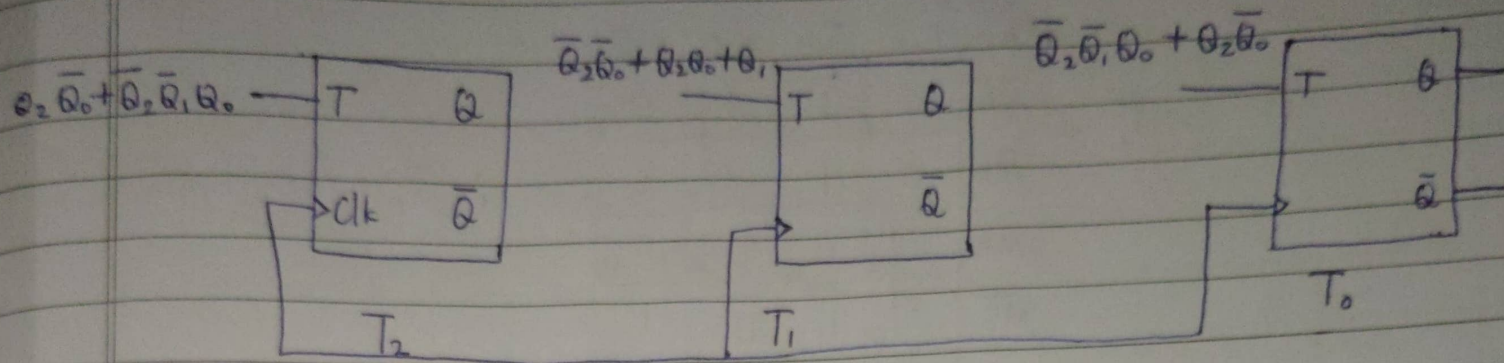
	$Q_1 Q_0$	00	01	11	10
Q_2	$\bar{Q}_2$	1 ₀	0 ₁	1 ₃	1 ₂
Q_2	Q_2	0 ₄	1 ₅	X ₇	X ₆

$$T_1 = \bar{Q}_2 \bar{Q}_0 + Q_2 Q_0 + Q_1$$

$T_0 =$

	$Q_1 Q_0$	00	01	11	10
Q_2	$\bar{Q}_2$	0 ₀	1 ₁	0 ₃	0 ₂
Q_2	Q_2	1 ₄	0 ₅	X ₇	X ₆

$$T_0 = \bar{Q}_2 \bar{Q}_1 Q_0 + Q_2 \bar{Q}_0$$



2) Design a 3 bit up/down counter using negative edge triggered T-FF.

⇒ Input M	Present state $Q_2 \quad Q_1 \quad Q_0$			Next state $Q_2^+ \quad Q_1^+ \quad Q_0^+$			T flip-flop - input $T_2 \quad T_1 \quad T_0$		
	Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	T_2	T_1	T_0
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	1	0	1	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	1	1	1	1	1
1	1	0	1	1	0	0	0	0	1
1	1	1	0	1	0	1	0	1	1
1	1	1	1	1	1	0	0	0	1

$$\therefore T_0 = 1$$

$T_1 =$

	$A, \bar{A}$	$\bar{A}, B$	$\bar{A}, \bar{B}$	A, B	$A, \bar{B}$
$M\bar{A}_2$	0	1	1	0	2
$\bar{M}\bar{A}_2$	0	1	1	0	6
MA_2	1	0	0	1	14
$\bar{M}A_2$	1	0	0	1	10

$$\therefore T_1 = \overline{M}\theta_0 + M\overline{\theta}_0$$

$T_2 =$

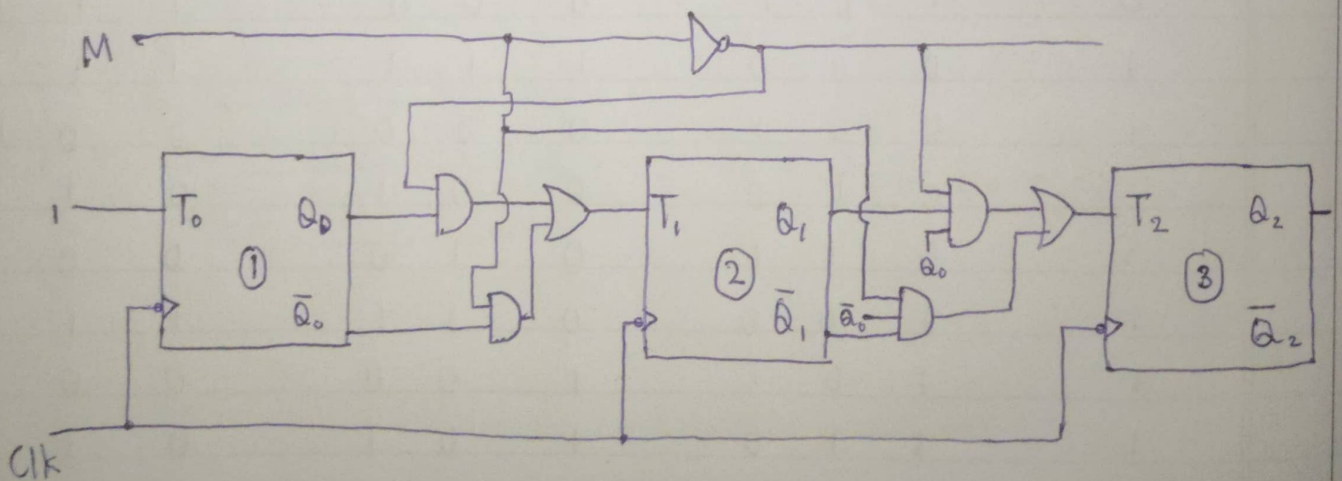
	$Q_1 \bar{Q}_0$	$\bar{Q}_1 \bar{Q}_0$	$Q_1 Q_0$	$\bar{Q}_1 Q_0$
$M \bar{Q}_2$	0	0	1	0
$\bar{M} \bar{Q}_2$	0	0	1	0
$M Q_2$	1	0	0	0
$\bar{M} Q_2$	1	0	0	0

$$\therefore T_2 = \bar{M}\theta_1\theta_0 + M\bar{\theta}_1\bar{\theta}_0$$

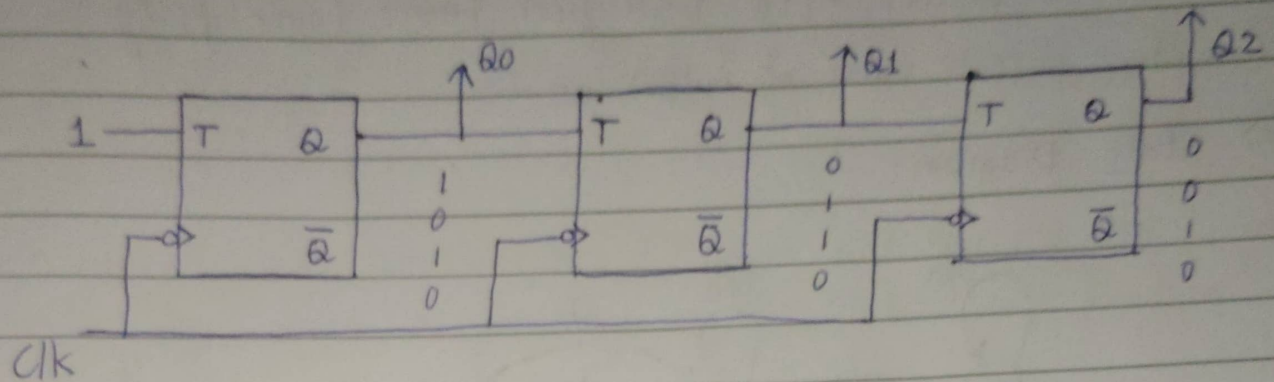
$$\therefore T_0 = 1$$

$$T_1 = \bar{M}Q_0 + M\bar{Q}_0$$

$$T_2 = \bar{M}Q_1Q_0 + M\bar{Q}_1\bar{Q}_0$$



- 3) The circuit in the figure below is a counter. What is the sequence that this circuit counts in?



$\Rightarrow$ These are negative edge triggered T-flip flops.

Assuming initial state $Q_0, Q_1, Q_2 = 000$.

Sequence	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	1	1	1
4	0	0	0
5	0	0	1
6	0	1	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$

$\therefore$ The sequence it follows is $000 \rightarrow 001 \rightarrow 010 \rightarrow 111 \rightarrow 000 \dots$